ENGR 1451/2451: Exploratory Data Science

Tests of Significance

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Today: Tests of Significance

Confidence Interval Review

A confidence interval estimates a population mean:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}.$$

A significance test asks whether the data are compatible with a specified value, such as $\mu = 70$.

- State a null and an alternative hypothesis.
- Measure a discrepancy in SE units.
- Calculate and interpret a P-value.
- Test a population mean or a success probability.

Starting Question: Is the Die Fair?

Recall the fair-die model

For one roll of a fair six-sided die,

$$\mu = 3.5, \qquad \sigma = \sqrt{35/12} \approx 1.708.$$

Now roll the die 100 times. The total number of spots is

$$368.$$

So the observed average is

$$\bar{x} = 3.68.$$

Question

Is a difference of 0.18 large enough to conclude that the die is not behaving like a fair die?

How Unusual Is 3.68 Average?

If the die is fair

For the average of $n = 100$ independent rolls,

$$\mathrm{SE}(\bar{X}) = \frac{1.708}{\sqrt{100}} \approx 0.171.$$

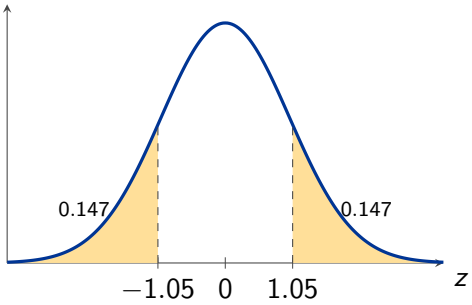
The observed average is therefore

$$z = \frac{3.68 - 3.50}{0.171} \approx 1.05.$$

Interpretation

The observed average is only about **1.05 SEs** above the fair-die value.

From z to P: Shade the Tail Area



Two-sided tail area

Under H_0, the P-value is the probability of being at least this far from 0 in *either* direction:

$$P = \Pr(|Z| \geq 1.05).$$

By symmetry,

$$P = 2\Pr(Z \geq 1.05) \approx 2(0.147) \approx 0.294.$$

Same Difference, More Data

Suppose instead the die is rolled **1,000 times** and the average is still 3.68 (total = 3680 spots).

The observed difference is unchanged

$$3.68 - 3.50 = 0.18.$$

But the SE is much smaller:

$$\mathrm{SE}(\bar{X}) = \frac{1.708}{\sqrt{1000}} \approx 0.054.$$

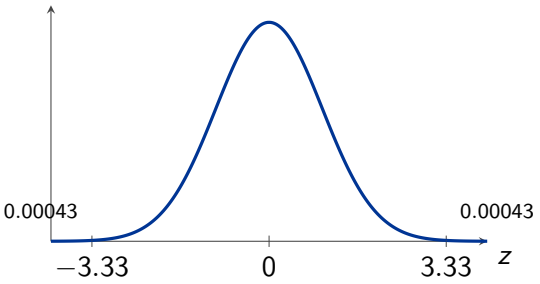
Now compare with chance variation

$$z = \frac{3.68 - 3.50}{0.054} \approx 3.33.$$

The observed average is now about **3.33 SEs** above the fair-die value.

Question: Why does the same difference of 0.18 provide much stronger evidence with 1,000 rolls than with 100 rolls?

From z to P: Much Smaller Tail Area



Two-sided tail area

For the 1,000-roll example,

$$z \approx 3.33.$$

Under H_0,

$$P = \Pr(|Z| \geq 3.33).$$

By symmetry,

$$\begin{aligned} P &= 2\Pr(Z \geq 3.33) \\ &\approx 2(0.00043) \\ &\approx 0.00086 \approx 0.0009. \end{aligned}$$

The same observed difference is much less plausible under H_0 when the SE is smaller.

Is the Material Meeting Its Strength Target?

Quality-control test

A manufacturer claims that the mean tensile strength of a polymer is

$$\mu = 50 \text{ MPa}.$$

An engineer tests a random sample of 500 specimens and obtains

$$n = 500, \bar{x} = 48 \text{ MPa}, \text{SD} = 15.3 \text{ MPa}.$$

Question

Does the sample provide convincing evidence that the **true mean tensile strength is below 50 MPa**?

A. No. A difference of 2 MPa is small compared with an SD of 15.3 MPa.

B. Yes. A difference of 2 MPa is large compared with the SE of the sample mean.

The Null and the Alternative

For the opening example, suppose the research question is whether the population mean is *below* 50.

Null hypothesis

$$H_0 : \mu = 50$$

The sample mean varies around 50 under the sampling model.

Alternative hypothesis

$$H_A : \mu < 50$$

A population mean below 50 is the departure of interest.

Quick question: Does H_0 claim that the *sample* mean or the *population* mean equals 50?

The Test Statistic Measures the Discrepancy

Observed minus expected, in SE units

$$\text{test statistic} = \frac{\text{observed} - \text{expected under } H_0}{\text{SE of the observed statistic}}$$

Opening example: large-sample approximation

$$z \approx \frac{\bar{x} - \mu_0}{\widehat{\text{SE}}} = \frac{48 - 50}{15.3/\sqrt{500}} \approx -2.92.$$

- H_0 supplies the benchmark $\mu_0 = 50$.
- The sample supplies the estimated variability.
- Negative z means the observed mean is below the null benchmark.

Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), pp. 478–480..

Comparing Different Evidence

Predict first: which sample gives the strongest evidence?
Each sample has $\bar{x} = 48$. Test $H_0 : \mu = 50$ against $H_A : \mu < 50$.

Sample	n	SD	Estimated SE	z
Original	500	15.3	0.684	-2.92
More variable	500	33.6	1.503	-1.33
Smaller sample	100	15.3	1.530	-1.31

The P-Value Is a Tail Probability

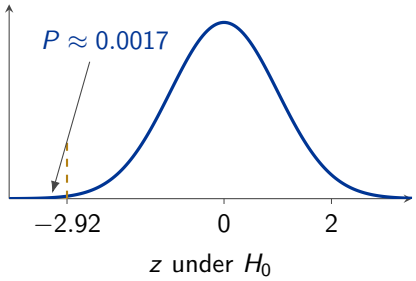
Assume the null model is correct

The P-value is the probability of a test statistic **as extreme as, or more extreme than,** the observed statistic in the direction of H_A.

For $H_A : \mu < 50$ and $z \approx -2.923$,

$$P \approx \Pr(Z \leq -2.923) \approx 0.0017.$$

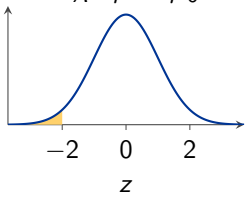
About 0.17% of statistics would be this low or lower under the normal approximation.



The Alternative Determines the Tail

Lower tail

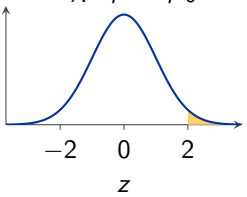
$H_A : \mu < \mu_0$



For $z = -2$, $P \approx 0.0228$.

Upper tail

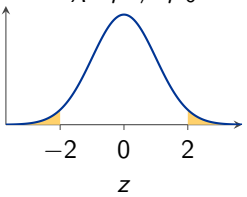
$H_A : \mu > \mu_0$



For $z = 2$, $P \approx 0.0228$.

Two tails

$H_A : \mu \neq \mu_0$



For $|z| = 2$, $P \approx 0.0455$.

Choose the alternative before examining the result.

What Does $P = 0.01$ Mean?

Vote true or false, then explain your reasoning
An upper-tail test gives $z = 2.35$ and $P \approx 0.01$.

1. If the null model is correct, about 1% of repeated samples give a statistic at least this large.
2. Given these data, the probability that H_0 is correct is 1%.
3. The probability that H_A is correct is 99%.

P and α Have Different Roles

Observed P-value

Calculated from the data using the null model.
Smaller P gives stronger evidence against that model.

Significance level α

A decision threshold chosen in advance, often $\alpha = 0.05$.
For a valid test, it controls how often a true null is rejected.

Decision rule	Interpretation
$P \leq \alpha$	Reject H_0 at that significance level.
$P > \alpha$	Fail to reject H_0; the evidence is insufficient.

Tail Probabilities in Python

Standard normal reference

```
model = stats.norm()
```

Student's t reference

```
model = stats.t(df=n-1)
```

```
from scipy import stats
```

Alternative / tail	Python probability
Lower tail	`model.cdf(statistic)`
Upper tail	`model.sf(statistic)`
Two tails, symmetric reference	`2 * model.sf(abs(statistic))`

Opening example: `stats.norm.cdf(-2.922965)` ≈ 0.001734.

SciPy reference: `scipy.stats.norm` and `scipy.stats.t`.

One-Sample t-Test with Unknown σ

Test $H_0 : \mu = \mu_0$ using the sample SD

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \mathrm{df} = n - 1$$

Why Student's t?

Estimating σ with s adds uncertainty. The t reference has heavier tails than the standard normal.

Conditions

Independent observations from a normal population give an exact t reference under H_0. Small samples make the normal-population assumption especially important.

At large n, t and normal reference curves become close.

Engineering Example: Is the Instrument Biased?

A reference gas contains exactly 70 ppm CO. Five readings are

78,	83,	68,	72,	88 ppm.

Assume independent normal measurement errors with unknown SD.

Scientific question

Does the instrument have systematic bias in either direction?

$$H_0 : \mu = 70, \qquad H_A : \mu \neq 70.$$

Before calculating

1. What statistic should be compared with 70?
2. Which SE should be used?
3. Which reference distribution?
4. How many degrees of freedom?

Calibration Result

For the five readings,

$$\bar{x} = 77.8, \qquad s \approx 8.07, \qquad \widehat{\mathrm{SE}} \approx 3.61.$$

Two-sided t-test

$$t = \frac{77.8 - 70}{3.61} \approx 2.16, \qquad df = 4$$

$$P \approx 0.097.$$

At $\alpha = 0.05$, fail to reject H_0.

95% confidence interval

$$77.8 \pm 2.776(3.61)$$

$$\approx [67.8,\ 87.8] \text{ ppm.}$$

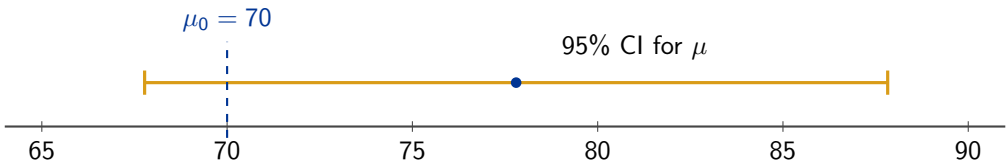
The interval includes 70 ppm.

For matching two-sided procedures, "70 is inside the 95% CI" and "$P > 0.05$" give the same decision.

Connection to a 95% Confidence Interval

Predict: What test decision follows from the interval? For the same five gas readings, the 95% confidence interval is

$$77.8 \pm 2.776(3.611) \approx [67.77,\ 87.83] \text{ ppm.}$$

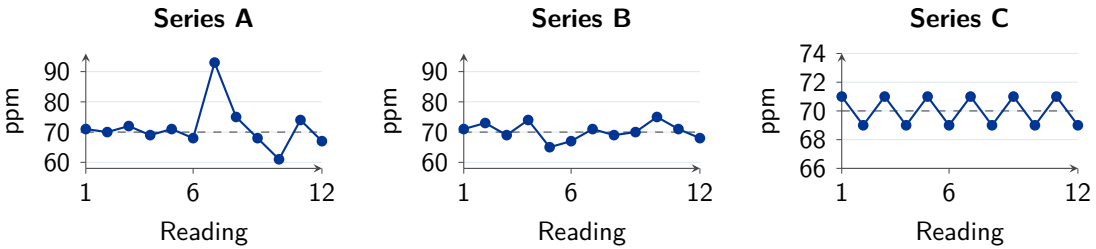


Question: Would a two-sided t-test reject $H_0 : \mu = 70$ at $\alpha = 0.05$?

Course extension connecting w07b to Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), span-gas data, p. 488..

Do the Measurement Assumptions Fit?

Discuss: Which two series raise concerns about independent normal errors? Each panel shows 12 successive measurements on the 70 ppm reference gas.



Data and question adapted from Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), Exercise F5, p. 494..

Summary: Interpreting a Test

1. **State the model and hypotheses** before examining the result.
2. **Standardize the discrepancy.**

$$\frac{\text{observed} - \text{expected under } H_0}{\text{appropriate SE}}$$

3. **Find the tail probability** using the reference distribution justified by the model.
4. **Report the estimated effect, uncertainty, and P-value** in context.

Interpretation to keep

A small P means the data are unusual under the specified null model. A large P leaves uncertainty. Practical importance and causal claims require additional reasoning.